

Ordinary Banach–Mazur rigidity from bounded Hochschild vanishing

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Abstract

Roydor’s cohomological rigidity theorem and the vanishing assertion in OpenAI’s family 295 manuscript imply local ordinary complex Banach–Mazur rigidity for every von Neumann algebra and its canonical predual. We give the precise composition, including an elementary adjoint argument for the predual conclusion. The threshold may depend on the algebra and the conclusion is Jordan $*$ -isomorphism. This draft records a conditional consequence, rather than a certified new unconditional theorem: Roydor’s actual author slides have been read, but his journal full text has not been obtained. The upstream proof has received independent mathematical audits; a fresh kernel build of its actual Lean declarations has not been reproduced. This document is not cleared for Zenodo publication as a full solution.

1 The target and conventions

All Banach spaces and linear maps are complex. For Banach spaces X, Y , put

$$d_{\text{BM}}(X, Y) = \inf_T \|T\| \|T^{-1}\|,$$

where the infimum ranges over bounded bijective complex-linear maps $T : X \rightarrow Y$ with bounded inverse. It is infinity when no such map exists. The literal norm-product formula gives $d_{\text{BM}}(0, 0) = 0$, and $d_{\text{BM}}(0, X) = d_{\text{BM}}(X, 0) = \infty$ for $X \neq 0$. For a von Neumann algebra M , M_* is its canonical Banach predual, with $M = (M_*)^*$ isometrically. The notation does not assume a trace or a semifinite realization.

A Jordan $*$ -isomorphism is a bijective complex-linear map preserving involution and $x \circ y = (xy + yx)/2$. On nonzero unital algebras it preserves the unit. The core question is whether, for each complex von Neumann algebra M , there exists $\varepsilon_M > 0$ such that for every complex von Neumann algebra N either inequality

$$d_{\text{BM}}(M, N) < 1 + \varepsilon_M, \quad d_{\text{BM}}(M_*, N_*) < 1 + \varepsilon_M \tag{1}$$

forces a Jordan $*$ -isomorphism. The comparisons are with von Neumann algebras and their canonical preduals. No assertion about arbitrary nearby Banach spaces is intended.

2 The two inputs

For $k \geq 1$, $C_b^k(M, M)$ denotes bounded complex k -linear maps with the usual multilinear operator norm; $C_b^0(M, M) = M$. For a k -cochain f , $k \geq 1$, the differential is

$$\begin{aligned} (df)(a_1, \dots, a_{k+1}) &= a_1 f(a_2, \dots, a_{k+1}) \\ &\quad + \sum_{j=1}^k (-1)^j f(a_1, \dots, a_j a_{j+1}, \dots, a_{k+1}) \\ &\quad + (-1)^{k+1} f(a_1, \dots, a_k) a_{k+1}. \end{aligned}$$

In degree zero, $(db)(a) = ab - ba$. We use

$$H_b^k(M, M) = \ker(d : C_b^k \rightarrow C_b^{k+1}) / \operatorname{im}(d : C_b^{k-1} \rightarrow C_b^k).$$

The denominator is the actual image, with no closure. Neither complete boundedness nor normality is imposed on these cochains.

Theorem 1 (Roydor’s reduction, author-slide formulation). *Let M be a complex von Neumann algebra with $H_b^2(M, M) = H_b^3(M, M) = 0$. There is $\varepsilon_M > 0$ such that, for every complex von Neumann algebra N , the following conditions are equivalent: M, N are Jordan $*$ -isomorphic; M, N are linearly isometric; M_*, N_* are linearly isometric; and each of the two strict distance inequalities in (1) holds.*

This is the stated scope of Theorem 2 in Roydor’s CIRM slides, interpreted in the standard complex operator-algebra setting, [2, PDF pp. 28–29], repeated later in the file. No separability, factoriality, type or trace hypothesis occurs in that statement. The bounded-cochain convention is introduced on PDF pp. 30–32 and distinguished from completely bounded cohomology. The displayed differential on p. 32 has apparent index/sign errors; we use the standard differential written above, also used by family 295. The published article is [1]; its precise theorem number, proof and correspondence with the slide version remain unverified because journal full-text access was unsuccessful. We therefore retain Theorem 1 as an explicitly identified external input rather than claiming to have independently reproved it. The slides do not explicitly declare the scalar field; the journal’s preliminary field convention remains to be checked.

Theorem 2 (OpenAI family 295 assertion). *For every complex von Neumann algebra M and every $k \geq 2$, every bounded complex k -linear Hochschild cocycle $M^k \rightarrow M$ has a bounded complex $(k - 1)$ -linear primitive. Thus $H_b^k(M, M) = 0$.*

This is Theorem 1.1 of the manuscript [3], pinned to repository commit `adc7f124` (the full identifier is recorded in the source receipt and the pinned link below). In particular, its assertion has exactly the field, coefficient module, boundedness, actual-image and degree conventions required above. Its proof reduces ordinary cocycles to separately normal cocycles, constructs uniformly bounded primitives on tracial type II_1 algebras using a unitary random walk and a Liouville argument, removes separability by conditional expectations and compactness, and assembles central summands with an explicit homotopy. The accompanying audit reports record which arguments were checked. This description is an audit summary, not a replacement proof of that input.

3 The consequence and an independent predual argument

Lemma 3. *For complex von Neumann algebras M, N ,*

$$d_{\text{BM}}(M, N) \leq d_{\text{BM}}(M_*, N_*).$$

Proof. If $T : M_* \rightarrow N_*$ is a bounded isomorphism, its Banach adjoint $T^* : N \rightarrow M$ is a bounded complex-linear isomorphism, with inverse $(T^{-1})^*$ and norms $\|T^*\| = \|T\|$, $\|(T^*)^{-1}\| = \|T^{-1}\|$. Inverting this algebra-space isomorphism does not change its distortion. Taking the infimum over T proves the inequality. If no T exists, the right side is infinite. Our zero convention gives the remaining cases. $\square$

Proposition 4 (Conditional composition). *If Theorems 1 and 2 hold in the stated scope, both targets in (1) hold for every complex von Neumann algebra, with a common algebra-dependent threshold. In that case Jordan $*$ -isomorphism, algebra-space linear isometry, predual linear isometry, and the two strict distance inequalities are equivalent.*

Proof. For $M \neq 0$, apply Theorem 2 in degrees 2 and 3, then apply Theorem 1. Its algebra threshold already suffices for the predual implication by Lemma 3. Thus the predual conclusion needs no quantitative estimate beyond the algebra reduction. The equivalences are supplied by the stated Roydor input. For $M = 0$, either strict distance inequality forces $N = 0$, so any positive threshold works. $\square$

There is no use of an infimum attaining map. If an ordinary distance is strictly below a given bound, some isomorphism has distortion strictly below that bound, directly from the definition of infimum. No continuity of the initial algebra-space isomorphism in a weak topology is assumed. The predual map's adjoint is weak-star continuous, but Roydor's ordinary-distance assertion does not require that extra property.

4 Scope, attribution and unresolved verification

The outcome sought is Jordan $*$ -isomorphism. Opposite algebras have the same underlying Banach norm and are Jordan $*$ -isomorphic, so ordinary Banach-space geometry cannot determine an orientation of the associative product. Likewise, no universal ε follows from the qualitative input: its quantifiers are $\forall M \exists \varepsilon_M > 0 \forall N$, not a uniform threshold over M .

Ricard and Roydor already proved unconditional stability for *completely bounded* Banach–Mazur distance [4, Theorem A]. Completely bounded distortion involves all matrix levels; its smallness is a stronger hypothesis than the ordinary distortion considered here. The earlier theorem therefore does not, by itself, supply the ordinary-distance conclusion. This draft does not claim a new completely bounded result.

Roydor's slides already state ordinary rigidity when the type II_1 summand has property Γ , a Cartan subalgebra, or McDuff absorption [2, PDF pp. 47–48]. Those subcases are inherited. Ricard–Roydor also record a conditional two-matrix-level distance version [4, Remark 3.7]; that stronger metric hypothesis is distinct from ordinary distance.

The claimed cohomology vanishing is attributed to OpenAI, and the cohomological geometric reduction to Roydor. The present composition is an immediate consequence if both dependencies are correct, not an independent solution of the Kadison–Ringrose cohomology problem or a new rigidity mechanism. The adjoint observation is standard Banach-space duality. A bounded current primary-source search and search of the pinned upstream manuscript corpus found no duplicate general ordinary consequence. Absence of a search result does not establish priority or firstness. The original reduction was publicly described before its journal publication; a 2020 slide date is not evidence for the upstream vanishing assertion's public priority.

The present package preserves a mathematical proof audit of family 295 and a static inspection of its genuine Lean declarations. It does not claim that a formalization catalogue or a comparator

stub verifies a theorem. A fresh Lean kernel reproduction remains pending, and the follow-on Banach–Mazur theorem is not formalized here. The independent mathematical source audit is the intended validation basis for the cohomology input; a missing kernel build is a limitation of formal evidence, not a mathematical counterexample. Full journal-source validation of Roydor’s reduction, completion of the priority audit, and fresh reviews of a final publication package are still required. No Zenodo deposit or tracker entry is warranted as a full resolution at this stage.

Disclosure. AI tools were used extensively for research, drafting and verification. Independent automated adversarial reviews are internal checks, not conventional human peer review. This research draft has not undergone conventional human peer review or refereeing.

References

- [1] J. Roydor, *Banach–Mazur stability of von Neumann algebras*, Journal of Topology and Analysis **14**(3) (2022), 767–792. [doi:10.1142/S1793525321500151](https://doi.org/10.1142/S1793525321500151). Published online December 11, 2020. Journal full text not read in this audit.
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- [3] OpenAI, *Vanishing of higher bounded Hochschild cohomology*, OpenAI Math Release preprint, manuscript dated September 23, 2026. [Pinned source and paper](#). Citation metadata supplied by the manuscript-specific README.
- [4] É. Ricard and J. Roydor, *A non-commutative Amir–Cambern theorem for von Neumann algebras and nuclear C^* -algebras*, Journal of Functional Analysis **267**(4) (2014), 1121–1136, [doi:10.1016/j.jfa.2014.05.018](https://doi.org/10.1016/j.jfa.2014.05.018); [arXiv:1108.1970v2](https://arxiv.org/abs/1108.1970v2).